

PHYS 143 – Mechanics and Thermal Physics

Temperature and Heat

What is temperature, how do we measure it, and how does thermal energy move?

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By the end of this week, you should be able to:

1. Distinguish temperature from heat, and explain thermal equilibrium and the zeroth law.
2. Convert between the Celsius, Fahrenheit and Kelvin scales and describe how thermometers work.
3. Calculate thermal expansion and the heat needed to change a temperature, including simple calorimetry.
4. Calculate the rate of heat transfer by conduction and radiation, and describe convection.

The detailed outcomes are in the checklist at the end of the lecture. **Reading:** Young and Freedman, Ch. 17, §17.1–17.4 and §17.6.

Three contacts:

1 DEFINE

What is temperature, and how do we measure it?

2 MODEL

What does a temperature change do, and what does it cost?

3 APPLY

How does heat move?

Practice: concept check → worked example
→ independent problem → application problem

Where we start: a five-minute check

Not marked. Answer individually, then we review.

1. Convert 2.5 kW to watts and 3.0 cm^2 to m^2 .
2. How much energy does a 100 W lamp use in 2.0 minutes?
3. A 2.0 m steel bar grows by 0.0010% . Find its length increase.
4. Solve $5(T - 32)/9 = 37$ for T .
5. Find the volume of a cube of side 0.10 m , in m^3 .
6. Two bodies exchange energy. If one gains 300 J , how much does the other lose?

1. $2500 \text{ W}; 3.0 \text{ cm}^2 = 3.0 \times 10^{-4} \text{ m}^2.$

2. $E = Pt = (100)(120) = 1.2 \times 10^4 \text{ J}.$

3. $\Delta L = 2.0 \times 10^{-5} \times 2.0 = 2.0 \times 10^{-5} \text{ m}.$

4. $T - 32 = 66.6 \Rightarrow T = 98.6.$

5. $(0.10)^3 = 1.0 \times 10^{-3} \text{ m}^3.$

6. It loses 300 J (energy is conserved).

Units and powers of ten (Lecture 1), energy and power (Lectures 4 and 5) are reused. Thermal physics adds one new quantity, temperature, and one new way energy is exchanged, heat.

Contact 1: Define

What is temperature, and how do we measure it?

Poll

A metal spoon and a wooden spoon have been in the same kitchen all day. You touch both. The metal one feels colder. **Which spoon has the lower temperature?**

Predict first

(a) the metal one, (b) the wooden one, (c) both are at the same temperature. What does your hand actually sense: the temperature of the spoon, or how fast energy leaves your hand?

Poll

A metal spoon and a wooden spoon have been in the same kitchen all day. You touch both. The metal one feels colder. **Which spoon has the lower temperature?**

Predict first

(a) the metal one, (b) the wooden one, (c) both are at the same temperature. What does your hand actually sense: the temperature of the spoon, or how fast energy leaves your hand?

(c). After a day in the same room both are in equilibrium with the air. Metal *conducts* energy away from your hand faster, so it *feels* colder. We will return to this in Contact 3.

Defining the vocabulary of temperature and heat

Quantity	What it is	What it measures	SI unit	Representation
Temperature T	the property of a body's state that decides which way energy flows	how hot or cold, on a chosen scale	K (also °C, °F)	a thermometer reading; a point on a scale
Heat Q	energy transferred between bodies because of a temperature difference	how much energy has moved	J	an arrow from the hotter to the cooler body
Thermal energy	the energy of the random motion of the particles	the energy stored inside a body	J	not drawn as a flow

Interpret: temperature answers *which way will energy flow?* Heat answers *how much energy moved?*

A body has temperature and thermal energy, but not “heat”.

- A **thermodynamic system** is the part of the universe we study; the rest is the **surroundings**.
- Its **state** is described by macroscopic variables: volume, pressure, mass, temperature.
- **Intensive** variables do not depend on the amount (temperature, pressure); **extensive** ones do (mass, volume).
- A **process** takes the system from one state to another.

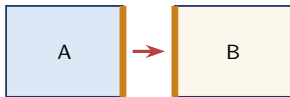
Physical meaning

We describe the system with a few measurable, large-scale quantities, without tracking individual molecules. Temperature is one of these.

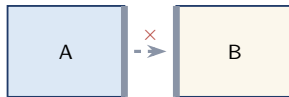
Predict first

Mass and temperature of a water sample: which is intensive? What happens to each if you split the sample in two?

Thermal equilibrium and walls



diathermal wall:
energy can pass



adiabatic wall:
no energy passes

Rule: thermal equilibrium

Two systems separated by a **diathermal** wall exchange energy until there is no *net* exchange: they are in **thermal equilibrium** and have the same temperature. An **adiabatic** wall prevents exchange.

Temperature is not heat

Temperature T

A property of the *state* of a body. It tells which way energy will flow when two bodies touch: from the higher to the lower temperature.

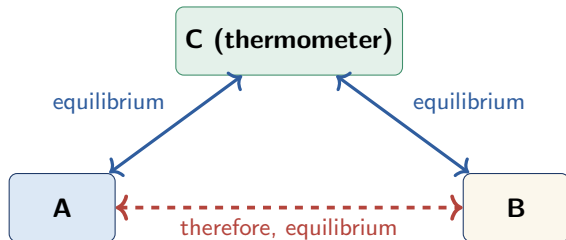
Heat Q

Energy *transferred* between bodies because of a temperature difference.
Unit: joule.

Common errors

- (1) “The body contains heat.” A body has thermal (internal) energy; heat is the energy *in transit*.
- (2) “Cold flows in.” Energy flows out of the warmer body.
- (3) “Equilibrium means no exchange.” It means no *net* exchange.

The zeroth law



Rule: the zeroth law

If A and B are each in thermal equilibrium with C , then A and B are in thermal equilibrium with each other.

Physical meaning

This is why thermometers work: C is the thermometer. If it reads the same when touching A and B , then A and B have the same temperature, even if they never touch.

How a thermometer works

- Choose a **thermometric property** that changes reproducibly with temperature.
- **Calibrate** it at fixed points (for example, melting ice and boiling water).
- Read the value when the thermometer is in equilibrium with the body.

Examples of thermometric properties:

- liquid in glass: length of the column;
- resistance thermometer: electrical resistance;
- constant-volume gas: gas pressure;
- infrared (temporal artery): radiated power.

Common error

A thermometer reads *its own* temperature. It is only meaningful once it has reached equilibrium with the body.

Decide first

1. Can energy flow between two bodies at the same temperature?
2. You place a thermometer in hot tea and it reads 80°C immediately. Is that the tea's temperature?
3. Does a body at a higher temperature always contain more thermal energy?
4. Can a temperature be lower than 0 K ?

Decide first

1. Can energy flow between two bodies at the same temperature?
2. You place a thermometer in hot tea and it reads 80°C immediately. Is that the tea's temperature?
3. Does a body at a higher temperature always contain more thermal energy?
4. Can a temperature be lower than 0 K ?

Answers

1. **No net flow.** Equal temperatures mean thermal equilibrium.
2. **Not until it reaches equilibrium.** A thermometer reads its own temperature.
3. **No.** A bathtub of warm water holds more thermal energy than a hot cup: energy also depends on the amount.
4. **No.** 0 K is the lower limit of the scale.

Temperature scales

Celsius	Fahrenheit	Kelvin	
100	-212	-373.15	steam point
37	-98.6	-310.15	body
0	-32	-273.15	ice point

Rule: scale conversions

$$T_F = \frac{9}{5}T_C + 32 \quad T_C = \frac{5}{9}(T_F - 32)$$

$$T_K = T_C + 273.15$$

Absolute zero, $0\text{ K} = -273.15^\circ\text{C}$, is the lower limit: a constant-volume gas thermometer extrapolates to zero pressure there.

EQUATION – Celsius and Fahrenheit

$$T_F = \frac{9}{5}T_C + 32$$

Meaning. the same temperature on two scales, with different degree sizes and zero points.

Variables. T_C in $^{\circ}\text{C}$; T_F in $^{\circ}\text{F}$.

Units. $^{\circ}\text{C}$, $^{\circ}\text{F}$.

Interpretation. a change of 1°C is 1.8°F ; the 32 applies to temperatures, not differences.

Example. $37.0^{\circ}\text{C} = 98.6^{\circ}\text{F}$.

EQUATION – kelvin

$$T_K = T_C + 273.15$$

Meaning. the absolute scale, zero at the lowest possible temperature.

Variables. T_C in $^{\circ}\text{C}$.

Units. kelvin, K (no degree sign).

Interpretation. same degree size as $^{\circ}\text{C}$, so $\Delta T_K = \Delta T_C$.

Example. $37.0^{\circ}\text{C} = 310.15\text{ K}$.

TOOLKIT 1 – Temperature scales and conversions

1. **IDENTIFY:** is the quantity a *temperature* T or a temperature *difference* ΔT ?
 2. **PREDICT** the size: a body at 37°C is about 98°F and 310 K .
 3. **CONVERT a temperature:** use the formulas above, in the right order.
 4. **CONVERT a difference:** $\Delta T_K = \Delta T_C$ and $\Delta T_F = \frac{9}{5}\Delta T_C$; *no* offsets of 32 or 273.15.
- RULE:** use kelvin in any formula with T^4 or a ratio of temperatures; $^{\circ}\text{C}$ is fine for ΔT .
- WATCH OUT:** adding 273.15 to a difference, or 32 to a Fahrenheit difference.
- CHECK:** is $T_K > 0$, and does the size agree with your prediction?

Example 1: body temperature

Question

Normal body temperature is 37.0°C . Express it in degrees Fahrenheit and in kelvin. A fever raises it to 39.0°C : express the *change* in K, in $^{\circ}\text{C}$ and in $^{\circ}\text{F}$.

Predict first

Is the change in kelvin larger, smaller or equal to the change in $^{\circ}\text{C}$? And in $^{\circ}\text{F}$?

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Predict first

Is the change in kelvin larger, smaller or equal to the change in $^{\circ}\text{C}$? And in $^{\circ}\text{F}$?

Temperature: $T_F = \frac{9}{5}(37.0) + 32 = 98.6^{\circ}\text{F}$; $T_K = 37.0 + 273.15 = 310.15\text{ K}$.

Change: $\Delta T = 2.0^{\circ}\text{C} = 2.0\text{ K}$ (equal, no offset); $\Delta T_F = \frac{9}{5}(2.0) = 3.6^{\circ}\text{F}$.

Evaluate: equal in K and $^{\circ}\text{C}$, larger in $^{\circ}\text{F}$, as predicted. A doctor who says “310 kelvin” is not alarming. ✓

Independent problem

- (a) An oven is set to 180°C . Express this in $^{\circ}\text{F}$ and in K .
- (b) Overnight the temperature falls from 12°C to -3°C . Find the change in K and in $^{\circ}\text{F}$.

Predict first

Is 180°C larger or smaller than 180°F ? Is ΔT in (b) positive or negative?

Independent problem

- (a) An oven is set to 180°C . Express this in $^{\circ}\text{F}$ and in K.
- (b) Overnight the temperature falls from 12°C to -3°C . Find the change in K and in $^{\circ}\text{F}$.

Predict first

Is 180°C larger or smaller than 180°F ? Is ΔT in (b) positive or negative?

(a) $T_F = \frac{9}{5}(180) + 32 = 356^{\circ}\text{F}$; $T_K = 180 + 273.15 = 453\text{ K}$.

(b) $\Delta T = -3 - 12 = -15^{\circ}\text{C} = -15\text{ K}$; $\Delta T_F = \frac{9}{5}(-15) = -27^{\circ}\text{F}$.

Evaluate: $180^{\circ}\text{C} > 180^{\circ}\text{F}$ (since $180^{\circ}\text{F} \approx 82^{\circ}\text{C}$); the change is negative, as predicted. ✓

Common mistakes and the correct habit

Mistake

Saying a body “contains heat” or that “cold flows in”.

Adding 273.15 or 32 to a temperature difference.

Reading a thermometer before equilibrium.

Saying equilibrium means no exchange.

Equating a higher temperature with more thermal energy.

Instead

Heat is energy in transit; energy flows out of the hotter body.

Differences have no offsets: $\Delta T_K = \Delta T_C$.

It reads its own temperature.

It means no *net* exchange.

Thermal energy also depends on the amount.

System and state



Thermal equilibrium, zeroth law



Thermometers and scales

You should be able to

- distinguish temperature from heat;
- state the zeroth law and explain a thermometer;
- convert temperatures and temperature differences;
- explain absolute zero.

Connect: Lectures 10 and 11

Temperature as a measure of molecular kinetic energy (Lecture 10); the absolute scale in the gas laws.

Contact 2: Model

What does a temperature change do, and what does it cost?

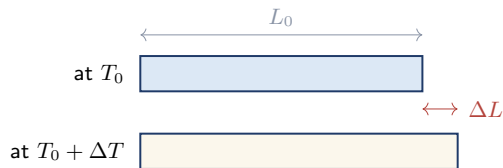
An opening question: gaps in the railway

Poll

Railway tracks and bridges have small gaps (expansion joints). What is the gap for, and how large must it be for a 12 m steel rail whose temperature varies by 40 K?

Predict first

Order of magnitude: 0.1 mm, 5 mm or 50 mm?



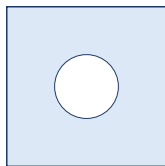
Rule: linear expansion

$$\Delta L = \alpha L_0 \Delta T$$

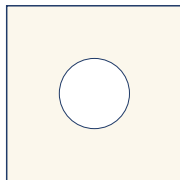
α : coefficient of linear expansion
(unit K^{-1}).

- Area: $\Delta A \approx 2\alpha A_0 \Delta T$. Volume: $\Delta V = \beta V_0 \Delta T$ with $\beta \approx 3\alpha$ for a solid.
- Valid for **small** ΔT and a uniform material; α depends slightly on temperature.

Does a hole expand or shrink?



cold plate

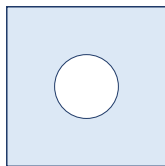


heated plate: hole grows too

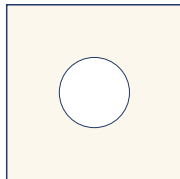
Predict first

A metal plate with a circular hole is heated. Does the hole get smaller, larger, or stay the same?

Does a hole expand or shrink?



cold plate



heated plate: hole grows too

Predict first

A metal plate with a circular hole is heated. Does the hole get smaller, larger, or stay the same?

Larger. Every length in the plate, including the diameter of the hole, scales by the same factor: the hole expands as if it were made of the same material. **Common error:** “the metal grows into the hole”.

EQUATION – linear expansion

$$\Delta L = \alpha L_0 \Delta T$$

Meaning. a solid lengthens in proportion to its length and the temperature change.

Variables. α coefficient of linear expansion; L_0 original length; $\Delta T = T_f - T_i$.

Units. α in K^{-1} ; ΔL in m.

Interpretation. positive for heating, negative for cooling; the fractional change is tiny.

Example. the 12.0 m rail: 5.8 mm.

EQUATION – volume expansion

$$\Delta V = \beta V_0 \Delta T, \quad \beta \approx 3\alpha$$

Meaning. the volume changes in proportion to the original volume.

Variables. β coefficient of volume expansion; V_0 original volume.

Units. β in K^{-1} ; ΔV in m^3 .

Interpretation. liquids usually expand more than solids.

Example. the fuel tank: 0.95 L of gasoline expansion.

TOOLKIT 2 – Thermal expansion

1. **IDENTIFY** the quantity that changes (length, area, volume) and the material (α from a table).
2. ΔT is final minus initial; the sign gives expansion (+) or contraction (–).
3. **CALCULATE:** $\Delta L = \alpha L_0 \Delta T$ (or $\Delta V = \beta V_0 \Delta T$, $\beta \approx 3\alpha$).
4. **COMPARE** with the size of the object (is $\Delta L/L_0$ small?).
5. **THERMAL STRESS:** if expansion is prevented, the stress is $F/A = Y\alpha\Delta T$ (Y = Young's modulus).

WATCH OUT: use L_0 at the starting temperature; hole dimensions scale like the solid.

CHECK: is the fractional change tiny (of order 10^{-4})?

Example 2: a steel rail

Question

A steel rail is 12.0 m long at 5°C. Find its length at 45°C ($\alpha = 1.2 \times 10^{-5} \text{ K}^{-1}$). What stress would develop if the rail could not expand ($Y = 2.0 \times 10^{11} \text{ Pa}$)?

Predict first

Expansion of the order of millimetres or centimetres?

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Predict first

Expansion of the order of millimetres or centimetres?

Set up: $\Delta T = 45 - 5 = 40 \text{ K}$.

Execute: $\Delta L = \alpha L_0 \Delta T = (1.2 \times 10^{-5})(12.0)(40) = 5.8 \times 10^{-3} \text{ m} = 5.8 \text{ mm}$.

Stress: $Y\alpha\Delta T = (2.0 \times 10^{11})(1.2 \times 10^{-5})(40) = 9.6 \times 10^7 \text{ Pa}$.

Evaluate: millimetres, as predicted ($\Delta L/L_0 = 4.8 \times 10^{-4}$, tiny); the stress is large, which is why joints are needed. ✓

Independent problem

A steel tank is filled to the brim with 50.0 L of gasoline ($\beta_{\text{gas}} = 9.5 \times 10^{-4} \text{ K}^{-1}$) at 10°C . The tank steel has $\beta_{\text{steel}} = 3.6 \times 10^{-5} \text{ K}^{-1}$. Estimate the volume that spills when it warms to 30°C .

Predict first

Which expands more, the gasoline or the tank? Does the spilled volume equal the gasoline's expansion alone?

Independent problem

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Predict first

Which expands more, the gasoline or the tank? Does the spilled volume equal the gasoline's expansion alone?

$\Delta T = 20 \text{ K}$. Gasoline: $9.5 \times 10^{-4} \times 50.0 \times 20 = 0.95 \text{ L}$. Tank:
 $3.6 \times 10^{-5} \times 50.0 \times 20 = 0.036 \text{ L}$.

Spill = $0.95 - 0.04 = 0.91 \text{ L}$. **Evaluate:** gasoline expands far more than the tank, as predicted. ✓

Concept check: heat and temperature change before the algebra

Decide first

1. Equal heat is given to 1 kg of water and 1 kg of copper. Which warms more?
2. A larger object expands more than a smaller one of the same material. Why?
3. Can an object absorb heat without its temperature rising?

Concept check: heat and temperature change before the algebra

Decide first

1. Equal heat is given to 1 kg of water and 1 kg of copper. Which warms more?
2. A larger object expands more than a smaller one of the same material. Why?
3. Can an object absorb heat without its temperature rising?

Answers

1. **The copper.** Its specific heat is about ten times smaller.
2. **More material, more length to expand:** $\Delta L \propto L_0$.
3. **Yes.** During a phase change (next lecture) the energy goes into breaking bonds.

Defining specific heat and heat capacity

Quantity	What it is	What it measures	SI unit	Representation
Specific heat c	the energy needed to raise 1 kg of a material by 1 K	how much energy a material absorbs per kelvin	J/(kg K)	a table value for each material
Heat capacity C	mc , the energy to raise the whole object by 1 K	how much energy a particular object absorbs per kelvin	J/K	a number for the object
Molar heat capacity C_m	the energy to raise one mole by 1 K	the energy per mole per kelvin	J/(mol K)	a table value; $C_m = Mc$

Interpret: specific heat answers *how much energy does this material need per kelvin?* Water's large c means it warms slowly and stores much energy.

Rule: heat and specific heat

$$Q = mc \Delta T \quad \text{unit of } c : \text{ J/(kg K)}$$

$Q > 0$: energy transferred *into* the body (its temperature rises). $Q < 0$: energy transferred *out*.

- Specific heat c : the energy per kilogram per kelvin; water: 4186 J/(kg K) .
- Heat capacity of an object: $C_{\text{obj}} = mc$.
- Molar heat capacity: $C = Mc$ (unit J/(mol K)).

Common error

A large c does not make something heat up quickly; it means a *lot* of energy is needed per kelvin. Water heats slowly.

EQUATION – heat for a temperature change

$$Q = mc \Delta T$$

Meaning. the energy transferred to change the temperature of a body without a phase change.

Variables. m mass; c specific heat;
 $\Delta T = T_f - T_i$.

Units. $\text{kg} \cdot \text{J}/(\text{kg K}) \cdot \text{K} = \text{J}$.

Interpretation. $Q > 0$ heating, $Q < 0$ cooling.

Example. the kettle: $5.0 \times 10^5 \text{ J}$.

EQUATION – calorimetry

$$\sum Q = 0$$

Meaning. energy lost by the hotter bodies equals energy gained by the cooler ones, in an isolated system.

Variables. $Q_i = m_i c_i (T_f - T_i)$ for each body, with signs.

Units. J.

Interpretation. the final temperature lies between the initial ones, nearer the larger heat capacity.

Example. copper in water: 22.9°C .

TOOLKIT 3 – Heat and specific heat

1. **IDENTIFY** the body, its mass m and material (c from a table).
 2. **PREDICT** the sign of Q : heating $\Rightarrow Q > 0$; cooling $\Rightarrow Q < 0$.
 3. $\Delta T = T_f - T_i$ (kelvin or $^{\circ}\text{C}$, the same size).
 4. **CALCULATE** $Q = mc\Delta T$ with m in kg and c in $\text{J}/(\text{kg K})$.
 5. **USE** $Q = C_{\text{obj}}\Delta T$ for an object, or $Q = nC\Delta T$ with the molar heat capacity.
- WATCH OUT:** this formula holds only if there is no phase change; c is assumed constant over the range.
- CHECK:** does the sign of Q agree with heating or cooling?

Example 3: heating water in a kettle

Question

A kettle heats 1.50 kg of water from 20°C to 100°C. (a) How much heat is needed? (b) How long does this take at a heating rate of 2.0 kW (all energy going into the water)?

Predict first

Order of magnitude of Q : 10^2 , 10^5 or 10^8 J? Minutes or hours?

Example 3: heating water in a kettle

Question

A kettle heats 1.50 kg of water from 20°C to 100°C. (a) How much heat is needed? (b) How long does this take at a heating rate of 2.0 kW (all energy going into the water)?

Predict first

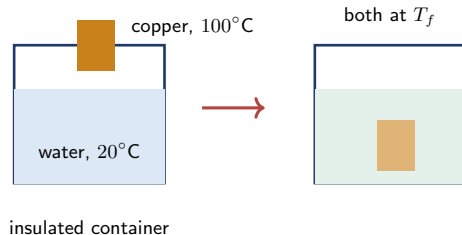
Order of magnitude of Q : 10^2 , 10^5 or 10^8 J? Minutes or hours?

(a) $Q = mc\Delta T = (1.50)(4186)(80) = 5.0 \times 10^5 \text{ J}$. (b)

$t = Q/P = 5.0 \times 10^5 / 2000 = 2.5 \times 10^2 \text{ s} \approx 4.2 \text{ min}$.

Evaluate: about 10^5 J and a few minutes, as predicted; $Q > 0$. ✓

Calorimetry: energy conservation for heat



Rule: calorimetry

In an isolated system, the heat gained by the cooler parts equals the heat lost by the hotter parts:

$$\sum Q = 0 \quad \text{or} \quad Q_{\text{gained}} = -Q_{\text{lost}}$$

TOOLKIT 4 – Calorimetry (no phase change)

1. **SYSTEM:** list every body that exchanges energy; insulate the system from the surroundings.
 2. **UNKNOWN** final temperature T_f (common to all bodies at equilibrium).
 3. **WRITE** $Q_i = m_i c_i (T_f - T_{i,\text{initial}})$ for each body, with signs.
 4. **SET** $\sum Q_i = 0$ and solve for T_f (a linear equation).
 5. **EVALUATE:** T_f must lie between the lowest and highest initial temperatures.
- WATCH OUT:** if T_f would reach a melting or boiling point, a phase change occurs: use the methods of the next lecture.
- CHECK:** does the heat lost by the hot body equal the heat gained by the cold one?

Example 4: a hot metal in water

Question

A 0.200 kg copper block ($c = 390 \text{ J}/(\text{kg K})$) at 100.0°C is dropped into 0.500 kg of water ($c = 4186 \text{ J}/(\text{kg K})$) at 20.0°C in an insulated container. Find the final temperature.

Predict first

Will T_f be closer to 100°C or to 20°C ? Why? (Compare the heat capacities.)

Example 4: a hot metal in water

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Predict first

Will T_f be closer to 100°C or to 20°C ? Why? (Compare the heat capacities.)

Execute: $m_{\text{Cu}}c_{\text{Cu}}(T_f - 100) + m_w c_w(T_f - 20) = 0$

$$(78.0)(T_f - 100) + (2093)(T_f - 20) = 0 \Rightarrow T_f = \frac{7800 + 41\,860}{78.0 + 2093} = 22.9^\circ\text{C}.$$

Evaluate: close to 20°C because the water's heat capacity (2093 J/K) is much larger than the copper's (78 J/K). Check: copper loses $78(77.1) = 6.0 \text{ kJ}$, water gains $2093(2.9) = 6.0 \text{ kJ}$. ✓

Independent problem

0.30 kg of water at 80°C is poured into 0.50 kg of water at 20°C in an insulated container. Find the final temperature.

Predict first

Will T_f be above or below 50°C (the simple average)? Why?

Independent problem

0.30 kg of water at 80°C is poured into 0.50 kg of water at 20°C in an insulated container. Find the final temperature.

Predict first

Will T_f be above or below 50°C (the simple average)? Why?

$$0.30 c (T_f - 80) + 0.50 c (T_f - 20) = 0 \Rightarrow T_f = \frac{0.30(80) + 0.50(20)}{0.80} = 42.5^\circ\text{C}.$$

Evaluate: below 50°C, as predicted, because there is more cold water than hot; between 20 and 80°C. ✓

Common mistakes and the correct habit

Mistake

Using $Q = mc\Delta T$ across a phase change.

Believing a large c means quick heating.

Thinking a hole shrinks when the plate is heated.

Taking L_0 at the final temperature.

Wrong signs in a calorimetry sum.

Instead

Use $\pm mL$ at the change (Lecture 10).

It means much energy per kelvin.

It expands like the material.

L_0 is the length at the starting temperature.

Write each $Q_i = m_i c_i (T_f - T_i)$ and set $\sum Q = 0$.

Temperature change



expansion: $\Delta L = \alpha L_0 \Delta T$

heat: $Q = mc\Delta T$



calorimetry: $\sum Q = 0$

You should be able to

- calculate linear, area and volume expansion;
- find the heat for a temperature change, with the correct sign;
- solve a calorimetry problem with no phase change.

Connect: Lectures 10 and 11

$Q = mc\Delta T$ leads to latent heat and phase changes (Lecture 10); heat and work are combined in the first law (Lecture 11).

Contact 3: Apply

How does heat move?

Three ways heat moves

Mechanism	How energy moves	Needs matter?	Example
conduction	collisions and vibrations between neighbouring particles, no bulk motion	yes (a solid, liquid or gas)	a spoon in hot tea
convection	the motion of the fluid itself carries the energy	yes (a fluid)	boiling water, sea breeze
radiation	electromagnetic waves	no	the Sun, a fire

Predict first

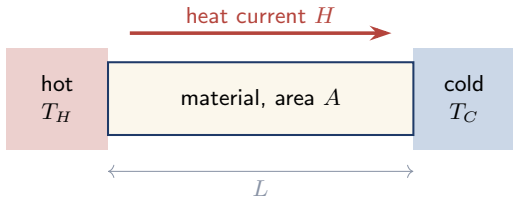
Which mechanism brings the Sun's energy to the Earth? Which warms your hand when you hold it beside a fire, not above it?

Defining the vocabulary of heat transfer

Quantity	What it is	What it measures	SI unit	Representation
Heat current H	the rate at which energy flows through a material or across a surface	how fast energy is transferred	W	an arrow from hot to cold
Thermal conductivity k	a material property in the conduction law	how well a material conducts energy	W/(m K)	a table value
Emissivity e	the ratio of the power radiated to that of an ideal radiator	how well a surface radiates	none	a number from 0 to 1

Interpret: the heat current answers *how fast does the energy move?* Conduction, convection and radiation are three ways to carry it.

Conduction



Rule: heat current in conduction

$$H = \frac{dQ}{dt} = kA \frac{T_H - T_C}{L}$$

k : thermal conductivity (W/(m K)).
 H in watts.

- Larger A or larger ΔT gives more; a longer path gives less.
- Metals have large k ; air, foam and wood have small k (insulators). Insulators *reduce* the rate; they do not stop the flow.

EQUATION – heat current in conduction

$$H = \frac{dQ}{dt} = kA \frac{T_H - T_C}{L}$$

Meaning. the steady rate at which energy flows through a slab from hot to cold.

Variables. k conductivity; A cross-section; $T_H - T_C$ temperature difference; L thickness.

Units. $\text{W}/(\text{m K}) \cdot \text{m}^2 \cdot \text{K}/\text{m} = \text{W}$.

Interpretation. proportional to A and ΔT , inversely proportional to L .

Example. the cooler lid: 20 W.

Predict first

Doubling the thickness of an insulating layer: predict what happens to the heat current before using the equation.

Predict the signs

A copper rod ($k = 385 \text{ W}/(\text{m K})$, $A = 2.0 \times 10^{-4} \text{ m}^2$) lies along x . The end at $x = 0$ is at 100°C and the end at $x = 0.50 \text{ m}$ is at 20°C . What is the sign of dT/dx , and in which direction does heat flow? What is H ?

Predict first

Hint: heat flows from hot to cold. Is the temperature rising or falling along $+x$?

Predict the signs

A copper rod ($k = 385 \text{ W}/(\text{m K})$, $A = 2.0 \times 10^{-4} \text{ m}^2$) lies along x . The end at $x = 0$ is at 100°C and the end at $x = 0.50 \text{ m}$ is at 20°C . What is the sign of dT/dx , and in which direction does heat flow? What is H ?

Predict first

Hint: heat flows from hot to cold. Is the temperature rising or falling along $+x$?

$$dT/dx = -\frac{80}{0.50} = -160 \text{ K/m} \text{ (negative: } T \text{ falls with } x\text{), so}$$

$H = -kA dT/dx = +(385)(2.0 \times 10^{-4})(160) = +12.3 \text{ W}$ along $+x$, from hot to cold. The minus sign is what makes the heat current point *down* the temperature gradient; it is not a negative conductivity.

But what if the hot end is at $x = 0.50 \text{ m}$? Then $dT/dx > 0$ and $H < 0$: the flow is along $-x$. The magnitude form $kA\Delta T/L$ gives the size only, not the direction.

TOOLKIT 5 – Conduction

1. **SKETCH** the hot side, the cold side, the path length L and the area A perpendicular to the flow.

2. **IDENTIFY** k and $\Delta T = T_H - T_C$ (in K or $^{\circ}\text{C}$).

3. **CALCULATE** $H = kA\Delta T/L$ (watts).

4. **SERIES:** when two layers carry the same current, set $H_1 = H_2$ and solve for the interface temperature.

WATCH OUT: L is the length *along* the flow, and A is the cross-section across it; the current is the same through every layer in steady state.

CHECK: is H positive from hot to cold, and is the interface temperature between T_C and T_H ?

Convert before you substitute

A window pane has area 1500 cm^2 and thickness 0.50 cm ($k = 0.80 \text{ W}/(\text{m K})$). The two faces differ by 10 K . Find H and the energy conducted in 30 min .

Predict first

Which quantities must be converted, and what must the unit of H be?

Convert before you substitute

A window pane has area 1500 cm^2 and thickness 0.50 cm ($k = 0.80 \text{ W}/(\text{m K})$). The two faces differ by 10 K . Find H and the energy conducted in 30 min .

Predict first

Which quantities must be converted, and what must the unit of H be?

$A = 0.150 \text{ m}^2$, $L = 5.0 \times 10^{-3} \text{ m}$, $t = 1800 \text{ s}$.

$H = kA\Delta T/L = (0.80)(0.150)(10)/(5.0 \times 10^{-3}) = 240 \text{ W}$ and $Q = Ht = 4.3 \times 10^5 \text{ J}$. A temperature *difference* needs no offset: $10 \text{ K} = 10^\circ\text{C}$.

But what if cm^2 and cm are used with k in $\text{W}/(\text{m K})$? Then $H = 24\,000 \text{ W}$, a hundred times too large: every length must be in metres.

Example 5: a cooler lid

Question

The foam lid of a cooler has area 0.60 m^2 and thickness 2.0 cm ($k = 0.033 \text{ W}/(\text{m K})$). The inside is at 5°C and the outside at 25°C . Find the heat current, and the energy conducted in one hour.

Predict first

Is the current of order 1 W , 20 W or 2000 W ?

Example 5: a cooler lid

Question

The foam lid of a cooler has area 0.60 m^2 and thickness 2.0 cm ($k = 0.033 \text{ W}/(\text{m K})$). The inside is at 5°C and the outside at 25°C . Find the heat current, and the energy conducted in one hour.

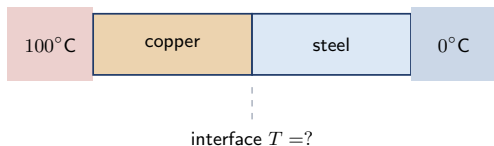
Predict first

Is the current of order 1 W, 20 W or 2000 W?

$$H = kA \frac{\Delta T}{L} = (0.033)(0.60) \frac{20}{0.020} = 20 \text{ W}. \quad Q = Ht = (20)(3600) = 7.1 \times 10^4 \text{ J per hour}.$$

Evaluate: about 20 W, as predicted: small, because k is small. This energy would warm 3.0 kg of drink by $\Delta T = Q/(mc) = 5.7 \text{ K}$ each hour. ✓

Example 6: two bars in series



A 0.30 m copper bar ($k = 385$) is joined end to end to a 0.30 m steel bar ($k = 50.2$), both of area 4.0 cm^2 . The ends are at 100°C and 0°C . Find the interface temperature and the heat current.

Predict first

Which bar has the larger temperature drop, copper or steel?

Two bars in series: solution

Set up: steady state: the same current in both: $H_{\text{Cu}} = H_{\text{steel}}$, and the lengths and areas are equal.

Two bars in series: solution

Set up: steady state: the same current in both: $H_{\text{Cu}} = H_{\text{steel}}$, and the lengths and areas are equal.

Execute: $385(100 - T) = 50.2(T - 0) \Rightarrow 38\,500 = 435.2T \Rightarrow T = 88.5^\circ\text{C}$.

$$H = \frac{kA(T - 0)}{L} = \frac{(50.2)(4.0 \times 10^{-4})(88.5)}{0.30} = 5.9 \text{ W}.$$

Two bars in series: solution

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$$H = \frac{kA(T - 0)}{L} = \frac{(50.2)(4.0 \times 10^{-4})(88.5)}{0.30} = 5.9 \text{ W}.$$

Evaluate: the interface is close to the copper's hot end: nearly all the temperature drop (88.5 K) is across the poor conductor (steel), as predicted. Check with copper:
 $(385)(4.0 \times 10^{-4})(11.5)/0.30 = 5.9 \text{ W}$. ✓

Concept check: why does metal feel colder?

Question

Back to the spoons: both are at 20°C , but your hand is at 33°C . Explain why the metal one feels colder.

Predict first

Which factor in $H = kA\Delta T/L$ differs between metal and wood?

Concept check: why does metal feel colder?

Question

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Predict first

Which factor in $H = kA\Delta T/L$ differs between metal and wood?

ΔT is the same for both, but $k_{\text{metal}} \gg k_{\text{wood}}$. The metal carries energy away from your skin much faster, so the skin cools more and the nerves report “colder”. The sensation measures the *rate* of energy transfer, not the temperature.

Decide first

A bottle wrapped in a wet cloth stays cooler than a dry one in a breeze. Does the water have to reach 100°C before it evaporates? Why does the evaporation cool the bottle?

Predict first

Hint: which molecules leave the surface, the faster or the slower ones?

Decide first

A bottle wrapped in a wet cloth stays cooler than a dry one in a breeze. Does the water have to reach 100°C before it evaporates? Why does the evaporation cool the bottle?

Predict first

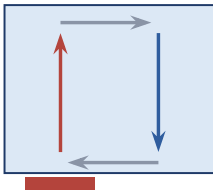
Hint: which molecules leave the surface, the faster or the slower ones?

No. Evaporation happens at any temperature (boiling is the special case of bulk vapour formation). The *faster* molecules escape, so the average kinetic energy of those that remain falls. The latent heat comes from the liquid, the cloth and the bottle. Size: evaporating 10 g takes about $(0.010)(2.4 \times 10^6) = 2.4 \times 10^4 \text{ J}$, enough to cool 1.0 kg of water by 5.8 K.

But what if the air is already saturated? Condensation then balances evaporation and there is no net cooling, which is why humid days feel hotter.

Convection

warm fluid rises, cool fluid sinks

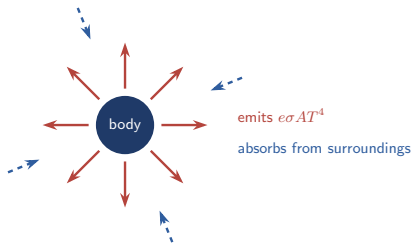


heat source below

Convection

Energy is carried by the *motion of a fluid*: warm, less dense fluid rises; cooler fluid sinks. Natural (buoyancy) or forced (fan, pump).

There is no single formula like conduction's, because the rate depends on the fluid's motion, the geometry of the surface, and the temperature difference together, and the motion is usually turbulent. Everyday examples: sea breeze, boiling, radiators, the body's blood flow.



Rule: radiation (Stefan–Boltzmann law)

$$H = e\sigma AT^4 \quad \sigma = 5.67 \times 10^{-8} \text{ W}/(\text{m}^2 \text{ K}^4)$$

e : emissivity ($0 \leq e \leq 1$); T in **kelvin**. Net exchange with surroundings at T_s :

$$H_{\text{net}} = e\sigma A (T^4 - T_s^4).$$

Every body radiates, whatever its temperature; a body at the same temperature as its surroundings still radiates, but absorbs just as much (thermal equilibrium).

EQUATION – Stefan–Boltzmann law

$$H = e\sigma AT^4, \quad H_{\text{net}} = e\sigma A(T^4 - T_s^4)$$

Meaning. the power a surface radiates; the net power when the surroundings radiate back.

Variables. e emissivity;
 $\sigma = 5.67 \times 10^{-8} \text{ W}/(\text{m}^2 \text{ K}^4)$; A area; T in *kelvin*.

Units. W.

Interpretation. doubling T multiplies H by 16; net loss only if $T > T_s$.

Example. a person: 870 W emitted, 140 W net.

Predict first

The absolute temperature of a hot object doubles.
Predict by what factor its radiated power changes.

TOOLKIT 6 – Radiation

1. **IDENTIFY** the surface area A , emissivity e and the temperatures of the body and surroundings.
 2. **CONVERT** all temperatures to **kelvin** (Toolkit 1).
 3. **EMITTED POWER:** $H = e\sigma AT^4$.
 4. **NET POWER** with surroundings at T_s : $H_{\text{net}} = e\sigma A(T^4 - T_s^4)$.
- WATCH OUT:** using $^{\circ}\text{C}$ in T^4 ; forgetting the radiation absorbed from the surroundings.
- CHECK:** is $H_{\text{net}} > 0$ when the body is hotter than its surroundings?

Example 7: radiation from a person

Question

A person has skin temperature 33°C , surface area 1.8 m^2 and $e = 0.97$. (a) What power do they radiate? (b) What is the net loss in a room at 20°C ?

Predict first

Order of magnitude: 10 , 10^2 or 10^4 W ? Is the net loss larger or smaller than (a)?

Example 7: radiation from a person

Question

A person has skin temperature 33°C , surface area 1.8 m^2 and $e = 0.97$. (a) What power do they radiate? (b) What is the net loss in a room at 20°C ?

Predict first

Order of magnitude: 10 , 10^2 or 10^4 W ? Is the net loss larger or smaller than (a)?

$$T = 306\text{ K}, T_s = 293\text{ K}.$$

$$(a) H = e\sigma AT^4 = (0.97)(5.67 \times 10^{-8})(1.8)(306)^4 = 8.7 \times 10^2\text{ W}.$$

$$(b) H_{\text{net}} = (0.97)(5.67 \times 10^{-8})(1.8)[306^4 - 293^4] = 1.4 \times 10^2\text{ W}.$$

Evaluate: the net loss is much smaller, because the person also absorbs from the room, and it is of order 10^2 W , comparable with the body's metabolic power. ✓

TOOLKIT 7 – Rates and energy

1. **RATE:** identify the heat current H (W) from conduction (Toolkit 5) or radiation (Toolkit 6).

2. **ENERGY:** $Q = Ht$ for a constant current over a time t .

3. **EFFECT:** link to the body: $Q = mc\Delta T$ (Toolkit 3), no phase change.

4. **SOLVE** for the unknown (t , ΔT or H).

WATCH OUT: H changes if T changes: use the initial rate for small changes only, and say so.

CHECK: are the units consistent: $\text{W} \times \text{s} = \text{J}$?

Lectures 10 and 11

Temperature will be linked to the average kinetic energy of molecules; heat Q and work W together change the thermal energy in the first law.

Lecture 12

Heat always flows from hot to cold: this one-way direction is the second law, and sets the limits of engines.

Applications

Insulation, the greenhouse effect, climate and the cooling of the body all use conduction, convection and radiation together.

Physical meaning

Calorimetry is energy conservation (Lecture 5) applied to heat: nothing is lost, only transferred.

Integrated problem: heating water through a rod

Application problem: decide what physics applies

A 0.50 kg sample of water at 20°C in an insulated container is heated by a copper rod ($k = 385 \text{ W}/(\text{m K})$, length 0.40 m, cross-section 2.0 cm^2) whose other end is in a reservoir at 100°C. Ignore losses.

- (a) What is the initial heat current?
- (b) About how long does the water take to warm by 1.0 K?
- (c) Is it reasonable to use the initial current for this estimate?

Predict first

IDENTIFY → mechanism and model → PREDICT → EXECUTE → EVALUATE.
Minutes or hours?

Identify, set up: conduction along the rod (Toolkit 5), then the effect on the water (Toolkit 3), linked by $Q = Ht$ (Toolkit 7). $A = 2.0 \times 10^{-4} \text{ m}^2$.

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$$(a) \ H = kA \frac{T_H - T_C}{L} = (385)(2.0 \times 10^{-4}) \frac{80}{0.40} = 15 \text{ W}.$$

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$$\text{(a)} \quad H = kA \frac{T_H - T_C}{L} = (385)(2.0 \times 10^{-4}) \frac{80}{0.40} = 15 \text{ W}.$$

$$\text{(b)} \quad Q = mc\Delta T = (0.50)(4186)(1.0) = 2.1 \times 10^3 \text{ J};$$
$$t = Q/H = 2093/15.4 = 1.4 \times 10^2 \text{ s} \approx 2.3 \text{ min}.$$

Identify, set up: conduction along the rod (Toolkit 5), then the effect on the water (Toolkit 3), linked by $Q = Ht$ (Toolkit 7). $A = 2.0 \times 10^{-4} \text{ m}^2$.

(a) $H = kA \frac{T_H - T_C}{L} = (385)(2.0 \times 10^{-4}) \frac{80}{0.40} = 15 \text{ W}.$

(b) $Q = mc\Delta T = (0.50)(4186)(1.0) = 2.1 \times 10^3 \text{ J};$

$t = Q/H = 2093/15.4 = 1.4 \times 10^2 \text{ s} \approx 2.3 \text{ min}.$

(c) Yes: the water warms by 1 K, so T_C changes from 20 to 21°C and ΔT from 80 to 79 K, a 1% change in H .

Evaluate: minutes, as predicted, for a small temperature change; warming all the way to 100°C would take hours, because the current falls as the water warms. ✓

Common mistakes and the correct habit

Mistake

Instead

Using $^{\circ}\text{C}$ in T^4 .

Kelvin for radiation and ratios.

Forgetting the radiation absorbed from the surroundings.

Use $e\sigma A(T^4 - T_s^4)$.

Using L across the flow in conduction.

L is along the flow; A is across it.

Confusing the heat current H with the energy Q .

$Q = Ht$.

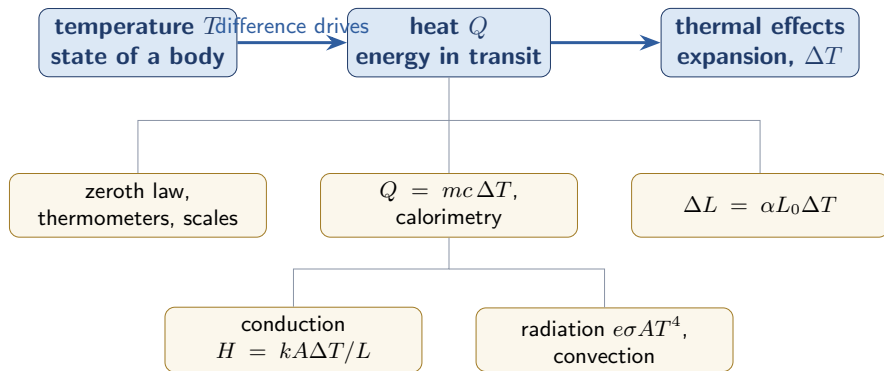
Saying insulators stop the flow.

They reduce the rate; they do not stop it.

Saying the metal spoon is at a lower temperature.

It conducts energy away faster, so it feels colder.

Lecture 9 concept map



MASTER PROCEDURE – THERMAL

1. **IDENTIFY** The system, the bodies and the process: what is asked?
 2. **SKETCH** The bodies, the boundaries and the direction of energy flow.
 3. **UNITS** Convert temperatures: kelvin for T^4 ; differences in K or $^{\circ}\text{C}$.
 4. **PREDICT** Sign of Q , direction of heat flow, and the rough size.
 5. **SELECT THE MODEL** Scale, expansion, specific heat, calorimetry, conduction or radiation.
 6. **EXECUTE** Write the equations first, then substitute with units.
 7. **EVALUATE** Signs, units, magnitude, limiting cases, and a second check (energy conservation).
- SKETCH $\rightarrow$ PREDICT $\rightarrow$ CALCULATE $\rightarrow$ CHECK.

Inside the course framework

IDENTIFY $\leftrightarrow$ steps 1–2
SET UP $\leftrightarrow$ steps 3–5
EXECUTE $\leftrightarrow$ step 6
EVALUATE $\leftrightarrow$ step 7

Physical meaning

Next lecture: what happens when the temperature reaches a melting or boiling point (phase changes and latent heat).

THE SEVEN TOOLKITS

- | | | |
|---|------------------------------------|---|
| 1 | Temperature scales and conversions | T against ΔT ; kelvin when needed |
| 2 | Thermal expansion | $\Delta L = \alpha L_0 \Delta T$; $\beta \approx 3\alpha$; thermal stress |
| 3 | Heat and specific heat | $Q = mc\Delta T$; sign; units |
| 4 | Calorimetry (no phase change) | $\sum Q = 0$; T_f between the initial values |
| 5 | Conduction | $H = kA\Delta T/L$; series layers |
| 6 | Radiation | $H = e\sigma AT^4$; net exchange; kelvin |
| 7 | Rates and energy | $Q = Ht$ combined with $Q = mc\Delta T$ |

The Master Procedure sits above these. Earlier toolkits (units, energy and power, signs) are reused, not renamed.

Key ideas

- A body has thermal energy, not heat.
- Use kelvin for T^4 and ratios; differences are equal in K and $^{\circ}\text{C}$.
- A hole expands with its material.
- Calorimetry is energy conservation: $\sum Q = 0$.
- Conduction rate $\propto A \Delta T / L$.
- Radiated power $\propto T^4$.

Where to look

Definitions: the vocabulary and equation-anatomy slides of each contact.

Common mistakes: the last slide of each contact.

Connections: the Connect boxes and slides.

Summary: core equations

Idea	Equation	Conditions and units
Scales	$T_F = \frac{9}{5}T_C + 32, \quad T_K = T_C + 273.15$	differences: no offsets
Expansion	$\Delta L = \alpha L_0 \Delta T, \quad \Delta V = \beta V_0 \Delta T$	$\beta \approx 3\alpha$; small ΔT
Heat	$Q = mc\Delta T$	no phase change; J
Calorimetry	$\sum Q = 0$	isolated system
Conduction	$H = kA(T_H - T_C)/L$	steady state; W
Radiation	$H = e\sigma AT^4, \quad H_{\text{net}} = e\sigma A(T^4 - T_s^4)$	kelvin; W
Energy from a rate	$Q = Ht$	constant H

Checklist: can I ...

- ☐ distinguish temperature from heat?
- ☐ state the zeroth law and explain a thermometer?
- ☐ convert temperatures and temperature differences?
- ☐ calculate linear, area and volume expansion?
- ☐ calculate a thermal stress?
- ☐ find $Q = mc\Delta T$ with the correct sign?
- ☐ solve a calorimetry problem without a phase change?
- ☐ use $H = kA\Delta T/L$, including layers in series?
- ☐ use $H = e\sigma AT^4$ with kelvin?
- ☐ describe convection?
- ☐ combine a rate with a time and $Q = mc\Delta T$?
- ☐ select the appropriate model in an unfamiliar problem?

After this week

Tutorial sheet: extra conversion, expansion, specific-heat, calorimetry, conduction, radiation and combined problems. Supplementary notes: the gas thermometer and absolute zero, thermal stress, thermal resistance, and blackbody radiation in outline.